

Total No. of Questions : 8]

SEAT No. :

PC2352

[Total No. of Pages : 3

[6354]-468

B.E. (Civil)

AIRPORT AND BRIDGE ENGINEERING
(2019 Pattern) (Semester - VII) (Elective - IV) (401004D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Draw neat diagrams wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of scientific calculators is allowed.

- Q1)** a) What is BIM (Building Information Modelling) for airport engineering? Explain in detail. [6]
- b) State and explain the factors affecting the design of Airport capacity. [6]
- c) What is airport drainage? What are the functions and basic requirements of airport drainage? [6]

OR

- Q2)** a) Write note on. [6]
- i) Augmented reality
 - ii) Virtual Reality
- b) Explain planning and design of airport pavement. [6]
- c) Discuss in brief the necessity of airport drainage. [6]

- Q3)** a) Describe the following terms with neat diagram. [6]
- i) Runway marking
 - ii) Taxiway marking
- b) Why lighting and marking of airport is required? Enlist parameters Considered for heliport planning. [6]
- c) Explain marking of heliport with neat sketch. [5]

OR

P.T.O.

Q4) a) Explain vertical take-off and landing (VTOL), short take-off and Landing (STOL). **[6]**

b) Explain with sketch various markings on runways. **[6]**

c) What is heliport? State the various helicopter characteristics. **[5]**

Q5) a) Define bridge? What are the characteristics of an ideal site for a bridge? **[6]**

b) Define following terms related to bridge. **[4]**

i) Effective span

ii) Freeboard

c) The catchment area of a stream is of sandy soil with thick vegetation Cover and the area of the catchment is 12000 hectares. The length of the Catchment is 25 km and the fall in level from the critical point to the Bridge site is 180 meters. Calculate the peak run off for designing the Bridge, if the severest storm as recorded yielded 18 cm for rainfall of 4 hours. Area factor = 0.70 and Coefficient to account for Losses due to absorption = 0.20 **[8]**

OR

Q6) a) Give the detail classification of bridge. **[4]**

b) What are the different component parts of bridge explain with typical Cross section of bridge. **[6]**

c) Calculate peak storm discharge from a catchment of 80,000 hectare The run off coefficient is 0.80 length of catchment is 65 km. the difference between the critical point and bridge is 150 m. The Rainfall in 8 hrs is 20 cm and area factor is 0.6 **[8]**

Q7) a) Explain with neat sketch the following; **[6]**

i) Bascule bridge

ii) Suspension bridge

b) Define culvert mention 2 types of Culvert, Describe any one culvert with neat sketch. **[6]**

c) State the purpose of providing bearing in bridges. Enlist different types of bearing. **[5]**

OR

Q8) a) Differentiate between temporary and permanent bridges with example. **[6]**

b) Discuss any three types of movable Span bridges. **[6]**

c) State the requirements of an ideal bridge bearings. **[5]**

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